Kai Qin TAN

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Data Scientist with a strong foundation in Banking, Actuarial Science and experience delivering measurable business impact through **advanced analytics**, **machine learning**, and **AI solutions** in **regulated financial environments**. Currently supporting **cards & payments business** in a **digital bank**, driving revenue growth, strategic decisioning, and operational efficiency through data science initiatives. Proven in **designing** and **developing** scalable, backend ML solutions end-to-end, and partnering with product, engineering, and business teams to **translate insights into action**. Technically proficient in **Python**, **ML/AI**, **Snowflake**, and cloud-based GenAI tools, with a strong focus on production-ready models and sustainable analytics.

EXPERIENCE

Data Science, Specialist | GXBank Berhad | Petaling Jaya, MY
September 2025 – Present

- Under the Grab Financial, focusing on **Regional Payments** and **Cards** for Singapore and Malaysia.
- Spearheaded the deployment of Digibank's **first Model Context Protocol** (MCP) server, scaling adoption across the 10-person regional analytics team and driving **full setup integration for GXBank**.
- Architected a **unified Snowflake MCP gateway** across Singapore and Malaysia, configuring **regional context files** and **reusable agentic skills** (interactive HTML dashboarding, automated and agentic exploratory data analysis) to streamline analytics workflows.
- Built a **hyper-personalized marketing profile engine** using multi-stage prompt chaining, incorporating context truncation and token optimization strategies to control LLM compute budgets to support targeting.
- **Credit Card Launch Analytics:** Led core analytics for GXS Singapore's initial Credit Card launch; designed onboarding funnel tracking, activation frameworks, and spend persona models to optimize launch conversion.
- **Funnel Optimization & Root-Cause Analysis:** Conducted diagnostic root-cause analysis on MSME and Singpass digital onboarding funnels, identifying drop-off friction points to improve conversion efficiency and user experience.
- Engineered a **LightGBM classification model** (AUC = 0.90) to profile **high-value corporate depositors** for GXS Business Banking, targeting relationship management and balance retention strategies.
- **Commercial Campaign Intelligence:** Directed end-to-end campaign analytics for GXS Operating Accounts and Business Loans, evaluating acquisition channels, drawdown behavior, and campaign ROI to guide executive decision-making.

Data Scientist | Sun Life Malaysia Assurance | Kuala Lumpur, MY
January 2024 – August 2025

- **Developed** machine learning models using **XGBoost** and **neural networks** to predict client payment defaults, achieving a **70% accuracy rate**, supporting client campaign initiatives.
- **Reengineered** a legacy **SAS e-miner Cross-sell model** in Python with **Scikit-Learn**, fully documented to meet internal compliance and audit standards, improving maintainability.
- **Built** an OCR-driven Python application to automate data extraction and analysis, **unlocking RM800k/month in new business opportunities**, a 30% extra sales opportunity.
- Led the implementation of **Anaconda Python** across analytics teams, eliminating ~**RM300k in annual SAS licensing costs**, and introducing **GitLab**, **Apache Airflow**, and **Linux** to the team.
- **Collaborated** with vendors on a GenAI proof-of-concept using Amazon Bedrock and Business Q, showcasing the potential of **LLMs trained on internal knowledge bases** to automate client queries and insights.

Intern, Strategic Analytics | Sun Life Malaysia Assurance | Kuala Lumpur, MY
October 2022 – January 2023

- Developed the Campaign Management module for an internal-use drag-and-drop platform using Flask framework and SQLite server.
- Led the backend team in A/B testing software logic using the tree-node concept.
- Constructed a regression duration predictor model for the data extraction process with 90% accuracy.

PROJECT

Client Next Best Action (NBA), Sun Life Malaysia Project

- Designed and developed a **client journey framework** tailored to distinct personas and backgrounds, supporting the company's Next Best Action strategy for **personalised client engagement**.
- Built a machine learning model to predict **clients at risk of missing upcoming payments**, aimed at improving customer retention, increasing policy persistency, and **reducing conservation costs**.
- Achieved **70% accuracy** in identifying high-risk clients; model implementation is projected to **reduce conservation costs by 30%** and enable targeted, cost-effective interventions.

Formula One Race Winner Prediction, Personal Project

- Developed a machine learning model to classify F1 drivers into podium, points, or non-scorer categories with **90% accuracy**, using historical performance data and driver-specific attributes.
- Engineered a **custom scoring algorithm** to quantify a driver's track-specific performance potential, incorporating contextual and personal data.
- Built and deployed a full-stack web application using **React and Flask**, integrating race calendars and driver profiles via external APIs, hosted as a PaaS solution.
- Demonstrated **end-to-end ownership**—from open-source data collection and feature engineering to model deployment and frontend integration.

EDUCATION

Masters of Data Science | Universiti Malaya (UM)
October 2026 – Present

Bachelor of Science in Actuarial Science with Distinction | Universiti Tunku Abdul Rahman (UTAR)
October 2020 - October 2023

- **CGPA**: 3.97/4.00
- **Awards**: MDT Innovations Top Graduate, Jobstreet Best Graduate (LKCFES), Little Rain Trust Best Graduate 2024, 11 times President List

- **SKILLS:** Python, Machine Learning, Natural Language Processing, MCP, Analytics & Insights, SQL, R, Flask, Linux RedHat, Solution Design, Dashboarding, LLM Orchestration Framework
- **TOOLS:** Tensorflow, Scikit-Learn, PyTorch, Anaconda, Snowflake, SAS, dbt, Amazon Bedrock, GitLab CI/CD, Excel VBA, Docker, Apache Airflow, Langfuse
- **LANGUAGES**: English (Full Working Proficiency), Malay (Working Proficiency), Mandarin (Native), Japanese (JLPT N4)

CREDENTIALS

- **AWS re/Start program Graduate** and an **AWS Certified Cloud Practitioner**.
- **Microsoft Data Science certificate** program graduate.
- Represented the AS course and gave a talk to the existing foundation students on the course and its structure.

REFERENCES

Adrian Iskandar, J.P. Morgan
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